

STEALTH DESIGN

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SUBMITTED TO : Harpreet Singh and Satwinder jit

1.DETECTION PRONE AREA :

1. Factors Affecting Reflection

Radar reflection from an aircraft mainly depends on:

- Metallic Cavities (engine inlet, ducts, weapon bays)
- Direct Exposure to Radar (surface facing radar source)
- Curvature and Orientation (smooth vs sharp surfaces)
- Corner Reflector Effect (90° surface junctions)
- Highly Conductive Metallic Parts (compressor blades, edges)

2. Ranked Detection-Prone Areas (Highest → Lowest)

The major radar detection-prone areas in an aircraft are ranked as:

1. **Engine Inlet**
2. **Compressor Blades**
3. **Weapon Bays (Open Condition)**
4. **Vertical Tail Edges**
5. **Wing Leading Edges**
6. **Cockpit Canopy**

3. Physical Reason Behind Each Area

A. Engine Inlet (Highest Reflection)

- Acts like a **metallic cavity or waveguide**
- Radar waves enter and reflect multiple times
- Produces very strong return signal



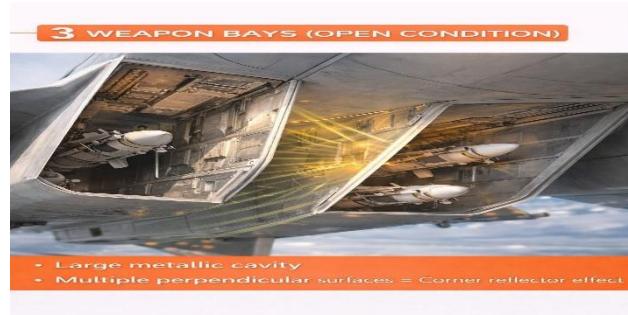
B. Compressor Blades

- Rotating metallic blades are highly reflective
- Curved surfaces cause strong specular reflections
- Even small exposure increases RCS sharply



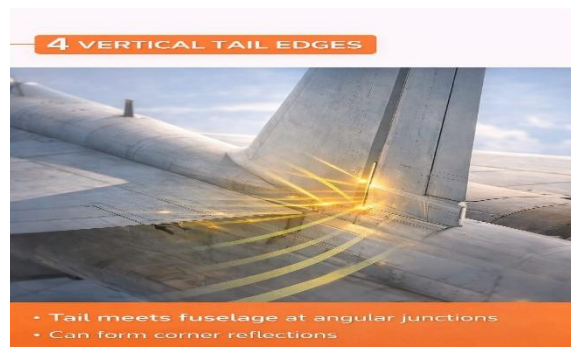
C. Weapon Bays (Open)

- Forms a large internal cavity
- Multiple perpendicular walls behave like a **corner reflector**
- Strong radar reflection when open



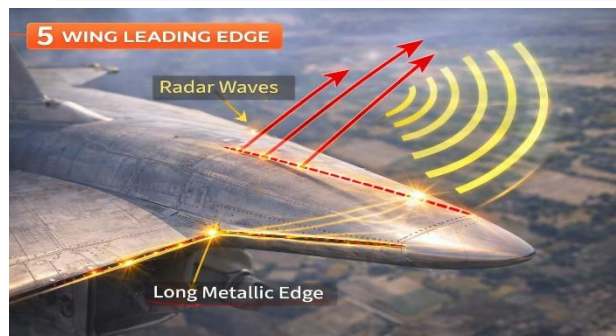
D. Vertical Tail Edges

- Tail meets fuselage at angular junctions
- Can create corner reflections
- Moderate-to-high radar return if surfaces are not angled



E. Wing Leading Edges

- Long metallic edges reflect radar strongly when perpendicular
- If aligned properly, energy gets deflected sideways reducing detection



F. Cockpit Canopy (Lowest Among These)

- Smooth rounded surface reduces direct reflection
- Often coated with thin metallic film
- Still slightly reflective but lowest in this ranking



2.DIFFERENT STRATEGIES FOR RADAR CONTROL :

Radar Detection Prone regions are aircraft surfaces that strongly reflect radar waves back to the receiver. To reduce radar visibility, multiple radar-reduction strategies are used. The major approaches and their possible implementation in this project are given below:

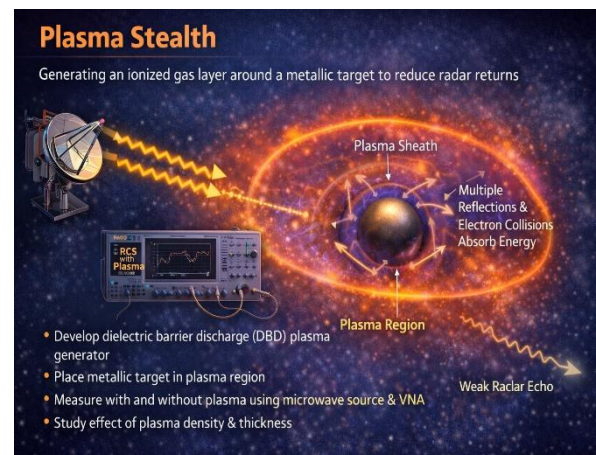
1. Plasma Stealth

What it is

Plasma stealth involves generating an ionized gas layer (plasma sheath) around a metallic surface using RF or electrical discharge.

How it reduces radar

Radar waves entering plasma undergo repeated reflections and energy absorption due to electron collisions, resulting in a weaker reflected echo.



2. Metamaterial-Based Stealth

What it is

Metamaterials are engineered surfaces with periodic unit cells (loops, patches, fractals) that manipulate electromagnetic waves.

How it reduces radar

Unit cells create phase cancellation or redirect reflected radar waves away from the radar source, lowering RCS.



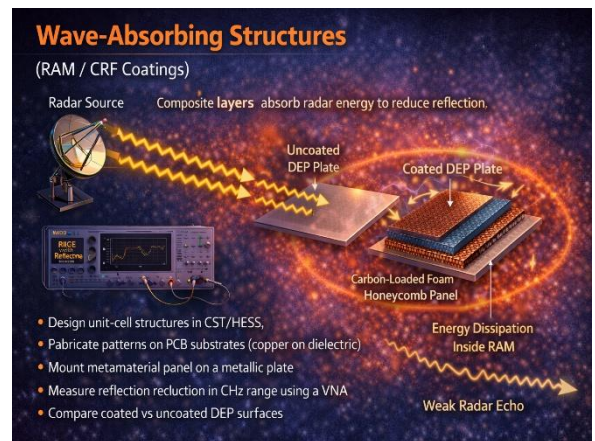
3. Wave-Absorbing Structures (RAM / CRF Coatings)

What it is

Radar Absorbing Materials (RAM) are composite layers (carbon-loaded foams, ferrite laminates, honeycomb panels) that absorb radar energy.

How it reduces radar

Radar waves enter the material instead of reflecting, and energy is dissipated internally as heat through dielectric and magnetic losses.



4. Diffusion of Radar Waves

What it is

Radar diffusion is the scattering of radar energy in many directions instead of reflecting it back directly.

How it reduces radar

Textured or angled surfaces spread radar returns spatially, so only a small fraction reaches the radar receiver.

Implementation in project

- Fabricate test panels with grooves, angled facets, or rough coatings.
- Use microwave scattering setup to observe reduced direct echo.
- Compare smooth vs diffusive surfaces experimentally.

5. Surface Roughness and Texturing

What it is :Surface texturing introduces controlled roughness in detection-prone aircraft zones.

How it reduces radar :Smooth surfaces produce strong specular reflection, while rough surfaces break coherence and scatter radar waves randomly.

Implementation in project

- Create textured test samples (etched or patterned metallic surfaces).
- Study radar reflection reduction compared to smooth plate.
- Identify DEP zones where texturing is most effective (leading edges, intake lips)

6. Fog / Aerosol Methodology

What it is

Artificial fog or aerosol clouds act as radar obscuration layers.

How it reduces radar

Suspended particles scatter and absorb radar waves, reducing signal strength and lowering detection probability.

Implementation in project

- Demonstrate attenuation using controlled aerosol/fog chamber.
- Transmit microwave signal through fog medium.
- Measure reduced signal-to-noise ratio at receiver.



7. Fibre Mat Radar Neutralization

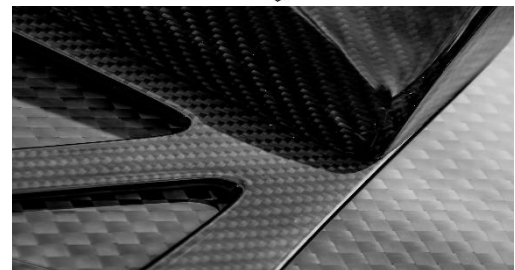
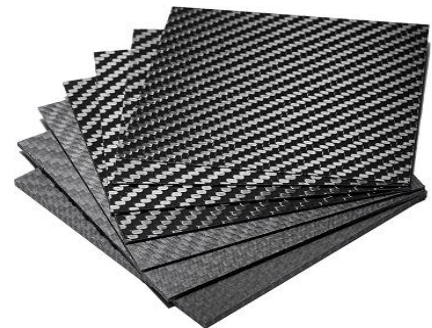
What it is Carbon fibre mats and conductive fibre composites act as electromagnetic damping layers.

How it reduces radar

Fibres introduce dielectric and conductive losses, dissipating radar energy inside the composite structure.

Implementation in project

- Use carbon fibre sheets as absorber layers on metallic targets.
- Measure reflection reduction using VNA.
- Evaluate lightweight structural stealth potential.



8. Quantum Communication (Low-Emission Optical Links)

What it is

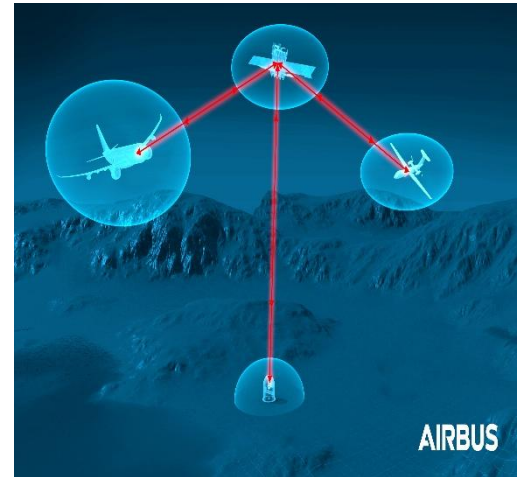
Quantum/optical communication uses faint photon-based transmission instead of RF signals.

How it reduces radar detectability

Optical links do not emit strong RF signatures, reducing electromagnetic detectability while maintaining secure communication.

Implementation in project

- Build a low-power laser communication demo link.
- Compare RF emissions of conventional transmitter vs optical system.
- Highlight reduced radar-visible electromagnetic footprint.



3. GEOMETRY-BASED RADAR CROSS SECTION (RCS) STUDY

1. What is Radar Cross Section (RCS)?

- Radar systems detect aircraft by transmitting electromagnetic (EM) waves.
- The reflected signal returning back to radar is quantified using **Radar Cross Section (RCS)**.

RCS indicates how strongly an object appears to radar.

- **High RCS** → Easily detectable
- **Low RCS** → Improved stealth

Units:

- m^2 (square meters)
- $dBsm$ (logarithmic radar signature scale)

2. Why Geometry Matters in Stealth

Stealth is not only achieved using absorbing materials, but also through **shape design**.

Geometric features affecting radar reflection:

- Flat metallic surfaces → Mirror-like strong reflection
- Sharp edges and corners → High scattering hotspots
- Smooth curved surfaces → Deflect radar waves away

Thus, aircraft geometry plays a major role in reducing radar detectability.

3. Aim of the Project

Main Objective:

- To study the effect of **surface curvature** (radius of curvature) on radar reflection and RCS.

Key focus:

- How changing curvature reduces backscatter
- Identification of detection-prone curvature ranges
- Development of geometry-based stealth guidelines

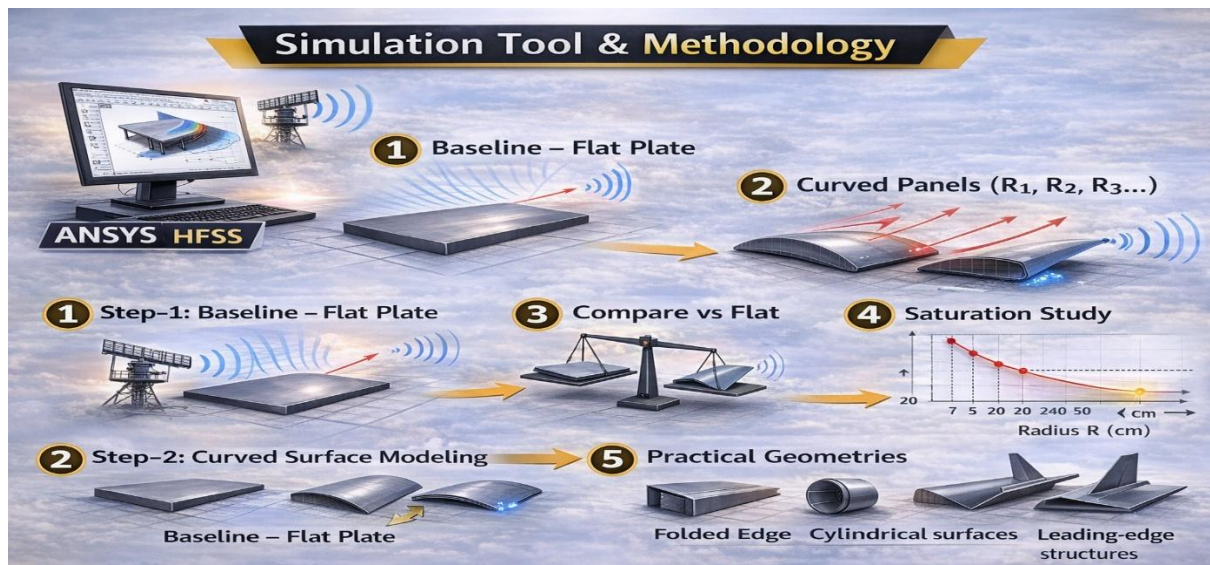
4. Simulation Tool and Approach

Software Used:

- **ANSYS HFSS (Electromagnetic Simulation Platform)**

HFSS will be used to compute RCS for different surface geometries under radar wave incidence.

5. Methodology and Workflow



Step 1: Baseline Reference Model

Start with a perfectly conducting **flat metallic plate**

Obtain reference RCS value

This serves as the comparison standard.

Step 2: Curved Surface Modeling

- Replace flat plate with a curved panel
- Define radius of curvature R

Parametric variation:

$$R = R_1, R_2, R_3, \dots$$

RCS is extracted for each curvature case.

Step 3: Comparative Analysis

Absolute RCS values are not sufficient, so results are interpreted using:

- Flat plate RCS (baseline)
- Curved surface RCS (modified case)

Performance metric:

$$\Delta RCS = RCS_{flat} - RCS_{curved}$$

This shows the improvement due to curvature.

Step 4: Saturation Study

- Radius is increased progressively
- Continue until a **saturation point** is reached

Saturation point:

- Further curvature increase gives negligible RCS reduction

Step 5: Extension to Practical Geometries

After curved panels, the study can be extended to:

- Folded edges
- Cylindrical surfaces
- Leading-edge profiles
- Junction-like structures

This reflects real aircraft zones with multiple local curvatures.

6. Expected Outcomes

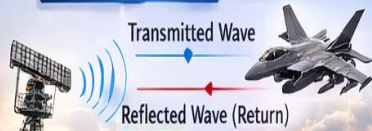
Final deliverables include:

- Curvature radius vs RCS trend plots
- Detection-prone curvature zones
- Curvature–RCS design grids
- Geometry-based guidelines for radar signature reduction

Geometry-Based Radar Cross Section (RCS) Study

Effect of **Surface Curvature** on Radar Reflection

1. What is RCS?

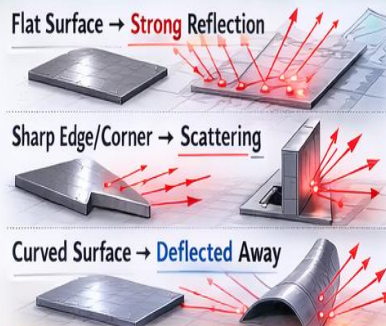


RCS = Measure of Radar Detectability

- High RCS → **Easy Detection**
- **Low RCS** → **Stealthy**

Units: m^2 , dBsm

2. Why Geometry Matters?



Goal → Minimize Reflected Energy Back to Radar

3. Aim of Project

Study Effect of Radius of Curvature (R) on **RCS**

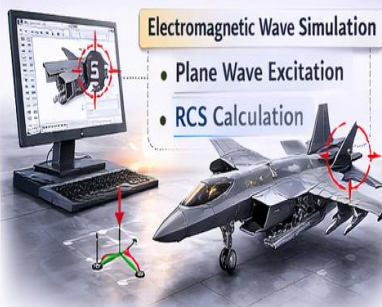
Objective:

- Reduce Radar Reflection
- Find **Optimal Curvature** for **Low RCS**

Small R → High RCS Large R → Low RCS

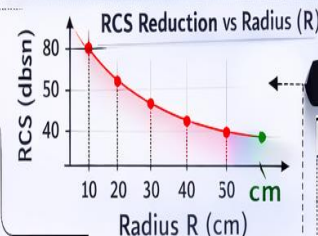


4. Simulation Tool: ANSYS HFSS

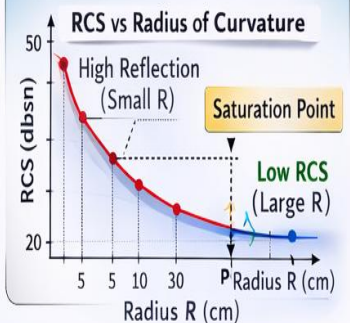


5. Methodology/Workflow

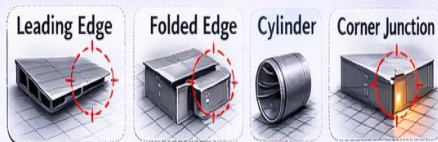
- Step-1: Flat Plate (Baseline) RCS = 10.5 dBsm
- Step-2: Curved Panels ($R_1, R_2, R_3, \dots$)
 $R = 1\text{ cm}, 5\text{ cm}, 10\text{ cm}, 20\text{ cm}$
- Step-3: Compare with Flat
 $\Delta \text{RCS} = \text{RCS}_{\text{flat}} - \text{RCS}_{\text{curved}}$



6. Results & Saturation



7. Extension to Practical Geometries



Design Guidelines for Stealth Aircraft

Final Outcome: Curvature-RCS Design Grid

Radius (R)	RCS (dBsm)	Reduction
Flat (0) →	10.5	—
1 cm →	8.7	↓ 1.8
5 cm →	5.2	↓ 5.3
10 cm →	3.1	↓ 7.4
20 cm →	2.8	↓ 7.7

Low RCS = **Stealthy**

Submitted By:

Name: Ayush Kumar Saw , Baljinder Kaur , Jasnoor Kaur , Kondala Seershika

Roll No: 2023meb1331 , 2023meb1334 , 2023meb1351 , 2023meb1355

Department: Mechanical Engineering

Institute: IIT Ropar